

Deepthi Nannapaneni

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Data Engineer/ Python Developer

- I am a highly experienced Data Engineer with around 5 years of experience in developing, implementing, and optimizing data plumbing systems and ETL processes.
- Successfully designed and implemented multiple projects through the entire development cycle, driving down storage cost and increasing customer satisfaction, streamlining integration, and profiling processes and more.
- Creating and executing test scripts, developing CI/CD pipelines, and engineering microservice-based architecture.
- Expertise in both relational databases and Big Data platforms, proficiency in Python and Spark programming, and a proven track record of success in leading and contributing to Big Data, Business Intelligence (BI), and Analytics projects.
- Demonstrated expertise in debugging and root cause analysis techniques, as well as implementing security frameworks for user and service authorization/authentication.

Skills & Competencies

- Cloud Computing (AWS, GCP)
- DevOps Methodologies
- Data pipeline design and implementation
- SQL and NoSQL Database Design and Development
- Big Data Technologies (Hadoop, Spark, Scala, Hive)
- Data Warehousing and Lake Solutions
- Data Modeling and Analysis
- Data Visualization tools (Tableau)
- Quality assurance/Data accuracy
- Data integration, profiling, and validation
- Database security
- ETL Development
- Data integration and API development
- Programming Languages (Python, Java, SQL)
- Automation Engineering
- Orchestration tools (Docker, Kubernetes).
- Operating system (Windows, LINUX, UNIX, MAC).
- Web Development (Django, Flask, HTML, CSS, JavaScript)

EDUCATION

Texas A&M University, Commerce- Computer & Information Sciences (Master of Science-3.51/4.00)

January 2022-May 2023

Coursework: Artificial Intelligence, Data Analysis and Visualization, Databases Design, Algorithms.

Jawaharlal Nehru Technological University, Kakinada (JNTUK)- Computer Science & Engineering (Bachelors – 3.4/4.00)

September 2014 – May 2018

Coursework: Java, Python, Database management system, Data structures & Algorithms, Computer Architecture, Web technologies.

Work Experience

Sr. Data Engineer/ Python Developer

Flying Frog Tech LLC

July 2023 -

Technology stack: Python, AWS, Django, Flask, Rest API, NumPy, Pandas, SQL, PostgreSQL, Spark, Jira, Git,

- Developed and maintained Python-based applications and scripts to support business operations and enhance efficiency.
- Utilized AWS services such as EC2, S3, and Lambda to design, deploy, and manage scalable and reliable cloud solutions.
- Implemented serverless architectures using AWS Lambda functions to optimize application performance and cost efficiency.
- Built RESTful APIs using Flask and Django frameworks to enable seamless integration with external systems.
- Implemented test cases using PyUnit, TDD, BDD approach to maintain code coverage.
- Experienced in building/maintaining Docker container clusters managed by Kubernetes, Linux, and Docker.
- Performed automation tasks on various Docker components like Docker Hub, Docker Engine, Docker Machine, Compose and Docker Registry.
- Working under UNIX environment in development of application using Python and familiar with all its commands.
- Collaborated with cross-functional teams to gather requirements, design solutions, and ensure timely project delivery.

- Managed and optimized relational databases (MySQL, PostgreSQL) on AWS RDS for data storage and retrieval.
- Developed a Tableau dashboard on top of the metric dataset to visualize performance and report to leadership team.
- Conducted code reviews, identified bugs, and implemented necessary fixes to maintain high code quality standards.
- Provided technical support and troubleshooting for applications running on AWS infrastructure.
- Used Git version control to manage the source code and integrated Git with Jenkins to support build automation and integrated with Jira to monitor the commits.

Data Engineering / Python Developer

Accenture - Verizon

August 2020 – December 2021

Technology Stack: Hadoop, GCP, Python, Django, SQL, Pandas, NumPy, Logging, GIT, Jenkins, Agile.

- Collaborated with Analytics and BI teams to develop globally adopted metrics and reporting-on-demand solutions, reducing manual data analysis by over 60%.
- Streamlined data integration, profiling, and validation for various data company-wide from multiple sources, increasing data accuracy by 40%.
- Developed and maintained stored procedures, views, and functions in SQL server to optimize data extract, transformation, and load (ETL) processes by 30%.
- Developed detailed data models and dictionaries for use in data warehouses, enabling stakeholders to easily access and report on critical organizational data.
- Involved in migrating on prem Hadoop system to using AWS.
- Managing and Scheduling Jobs on Hadoop cluster.
- Worked on analyzing Hadoop cluster and different big data analytic tools including Pig HBase database and Sqoop.
- Defining Automation Test Strategy, Test Approach and Test Planning for the Agile Sprints and releases.
- Knowledge of networking (TCP/IP, FTP, DNS, DHCP levels).
- Developed and maintained Verizon co-op crowd web application using Python, Django, and Flask, resulting in enhances performance and scalability.
- Created a report generation logic of the customers which saved a time to get the results of the customer.
- For the extraction of data from the database, we created queries and stored procedures for normalization and denormalization using MySQL.
- Followed Agile methodology to develop user stories, work through the backlog, and use Jira to document the stories and defect tracking.
- Collaborated with cross-functional teams to develop and maintain data governance policies and data security protocols, ensuring compliance with industry regulations and protecting sensitive data.

Data Engineer / Python Developer

Accenture - CIGNA

June 2018 – July 2020

Technology Stack: AWS RedShift, AWS S3 buckets, Glue, Lambda, SQL, pyspark scripts and Agile.

- Migrated data from AWS RDS SQL server to AWS RDS PostgreSQL.
- Developed Mappings to pull up the data into AWS S3 bucket from the SQL database.
- Built and maintained ETL pipeline to perform transformations like data pre-processing, aggregating tables and removing duplicate rows by writing spark scripts.
- Created mappings for dimensions using SCD type-2 logic which have source and target in AWS.
- created a logic for the complete data about source and target tables present in the database; this saved time and gave a complete list of those tables.
- Used Git as a source code management tool for performing functions such as branching, merging, and tagging etc.
- Generated dashboards and reports for business intelligence using AWS Quick Sight on data extracted from a PostgreSQL data warehouse.
- Troubleshoot and resolved the performance bottlenecks in ETL mappings and SQL overrides.
- Responsible for code build and promotion through different environments i.e., from Development to Production.
- Adhere to the Agile Methodology of Development that involves stand up/scrums meetings, sprint discussions and sprint retrospection.

Conference Submission

Submitted the paper titled " INTELLIGENT AI CHATBOT FOR ENHANCED CUSTOMER SERVICE " to the Hawaii International Conference on System Sciences (HICSS).

INTELLIGENT AI CHATBOT FOR ENHANCED CUSTOMER SERVICE

Technology stack: LSTM, integrating Dense, CNN, BI-LSTM features.

- Being able to assist consumers during their purchasing process and guarantee their pleasure depends on offering top-notch customer service.
- Customers expect businesses to provide a variety of channels for customer service, including social media, interactive voice response, websites, email, and live chat.
- These are the kinds of channels that should be available to customers. In this study, we show a customer service Chat-bot that uses a Deep Learning model with LSTM, integrating Dense, CNN, or BI-LSTM features, and improving accuracy up to 92%, exceeding existing methods.

PERSONALIZED CANCER DIAGNOSIS:

Technology stack: python, NLTK, matplotlib, sklearn, Machine learning Algorithms.

- We utilize textual clinical data to categorize specific genetic mutations into nine classes. Our objective is to predict data-points belonging to any of these nine classes.
- Developed a ML algorithm to predict the data-points which belong to any of the class which cause cancer.

ONLINE BOOKSTORE:

Technology stack: SQL, Visual Studio, XML.

- This project is to buy books online so they can easily find out whether the book is available or not, which will make them user-friendly.
- Customers can search for a book by title or author, add it to their shopping cart, and then pay with a credit card.